

In [2]:

```
!pip install opencv-python
```

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: opencv-python in .\AppData\Roaming\Python\Python314\site-packages (4.13.0.92)
Requirement already satisfied: numpy>=2 in .\AppData\Roaming\Python\Python314\site-packages (from opencv-python) (2.4.2)

In [3]:

```
!conda install -y opencv
```

'conda' is not recognized as an internal or external command,
operable program or batch file.

In [1]:

```
import cv2  
print(cv2.__version__)
```

ModuleNotFoundError Traceback (most recent call last)

Cell In[1], line 1
----> 1 import cv2
 2 print(cv2.__version__)

ModuleNotFoundError: No module named 'cv2'

In [2]:

```
import sys  
print(sys.executable)
```

C:\Users\afrah\AppData\Local\Python\pythoncore-3.12-64\python.exe

In [3]:

```
import sys  
!{sys.executable} -m pip install opencv-python
```

Collecting opencv-python

Using cached opencv_python-4.13.0.92-cp37-abi3-win_amd64.whl.metadata (20 kB)
Requirement already satisfied: numpy>=2 in .\AppData\Local\Python\pythoncore-3.12-64\Lib\site-packages (from opencv-python) (2.5.0)
Using cached opencv_python-4.13.0.92-cp37-abi3-win_amd64.whl (40.2 MB)
Installing collected packages: opencv-python
Successfully installed opencv-python-4.13.0.92

In [1]:

```
import numpy as np  
import matplotlib.pyplot as plt  
from tensorflow.keras.datasets import mnist, cifar10  
  
(mnist_x, _), _ = mnist.load_data()  
(cifar_x, _), _ = cifar10.load_data()  
  
print("Task 1: Pixel Values and Grayscale Calculation\n")  
  
for i in range(5):  
    r, g, b = cifar_x[i][16, 16]  
    gray = (int(r) + int(g) + int(b)) / 3  
    print(f"Image {i+1}: R={r}, G={g}, B={b} -> Gray = {gray:.2f}")
```

```

plt.figure(figsize=(12,5))

for i in range(5):
    plt.subplot(2,5,i+1)
    plt.imshow(mnist_x[i], cmap='gray')
    plt.title("MNIST")
    plt.axis("off")

for i in range(5):
    plt.subplot(2,5,i+6)
    plt.imshow(cifar_x[i])
    plt.title("CIFAR")
    plt.axis("off")

plt.tight_layout()
plt.show()

gray_images = []

for i in range(5):
    gray = np.mean(cifar_x[i], axis=2)
    gray_images.append(gray)

plt.figure(figsize=(12,4))

for i in range(5):
    plt.subplot(1,5,i+1)
    plt.imshow(gray_images[i], cmap='gray')
    plt.title("Grayscale")
    plt.axis("off")

plt.tight_layout()
plt.show()

```

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>

11490434/11490434 ————— 2s 0us/step

Downloading data from <https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz>

170498071/170498071 ————— 105s 1us/step

Task 1: Pixel Values and Grayscale Calculation

Image 1: R=245, G=237, B=226 -> Gray = 236.00

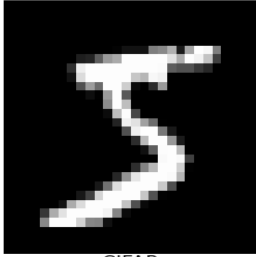
Image 2: R=218, G=221, B=216 -> Gray = 218.33

Image 3: R=112, G=100, B=68 -> Gray = 93.33

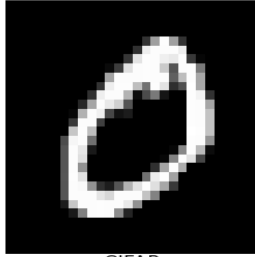
Image 4: R=181, G=144, B=109 -> Gray = 144.67

Image 5: R=92, G=100, B=113 -> Gray = 101.67

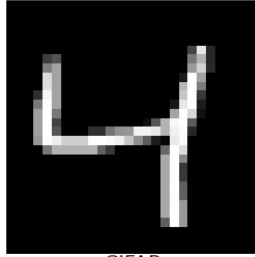
MNIST



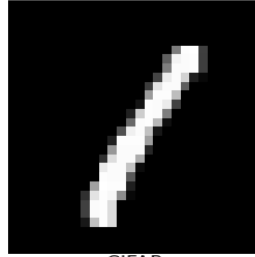
MNIST



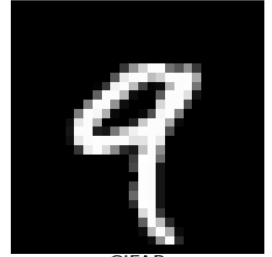
MNIST



MNIST



MNIST



CIFAR



CIFAR



CIFAR



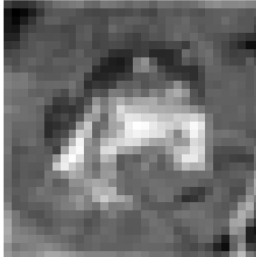
CIFAR



CIFAR



Grayscale



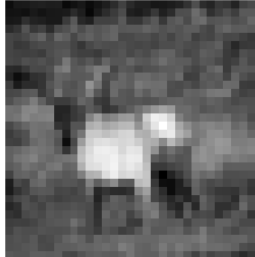
Grayscale



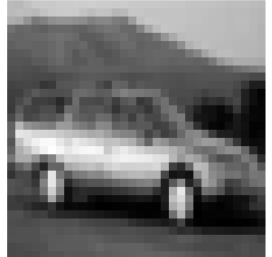
Grayscale



Grayscale



Grayscale



In []: